Definition

For all $x>0$, we define

$$\boxed{\int_1^x 1/t \, dt = \ln x} \quad **$$

$$\int_1^x t^2\,dt = \left(\frac{t^3}{3}\right)\Big|_1^x = \frac{x^3}{3} - \frac{1}{3}$$

- The expression $\ln x$ is called the natural logarithm of x.
- x has to be positive since $\int_1^x 1/t\,dt$ does not exist // when $x \le 0$.
- The FTC implies that

$$\frac{d}{dx}\left(\int_1^x \frac{1}{t}\,dt\right) = \frac{d}{dx}(\ln x)$$

$$\boxed{\frac{1}{x} = \frac{d}{dx}(\ln x)} \quad *$$

$$\frac{d}{dx}(\log_a x) = \frac{1}{x \ln a}$$

Let $a=e$.

$$\Downarrow$$

$$\frac{d}{dx}(\ln x) = 1/x$$

✓ We can deduce the following from **

① $\boxed{\ln 1 = 0}$

why?

$$\ln 1 = \int_1^1 \frac{1}{t}\,dt = 0$$

$$\log_e p = \ln p.$$

$$\log_a p^r = r\log_a p$$

② $\boxed{\ln e = 1}$

why?

$$\ln e^x = \int_1^{e^x} \frac{1}{t}\,dt$$

$$\ln e = \log_e e$$

$$= 1$$

$$x \ln e = \int_1^{e^x} \frac{1}{t}\,dt = x$$

$$x\ln e = x \iff \boxed{\ln e = 1}$$

$$y = \log_a p \iff p = a^y$$

Theorem

To every real number x, there's a unique (corresponding) positive real number y s.t.

$$x = \ln y$$
$$\Updownarrow$$
$$y = e^x$$

<u>Recall.</u>

The function g is an inverse of f iff

$$g(f(x)) = x \quad \text{and}$$
$$f(g(x)) = x$$

✓ We say that g and f are inverses of each other.

Definition

The natural exponential fn, denoted by exp is defined by e^x

$$e^{x} = y \iff \ln y = x \quad \forall x, \text{ where } y>0$$

$$e^{\ln y} = y \Leftarrow \ln y = x$$

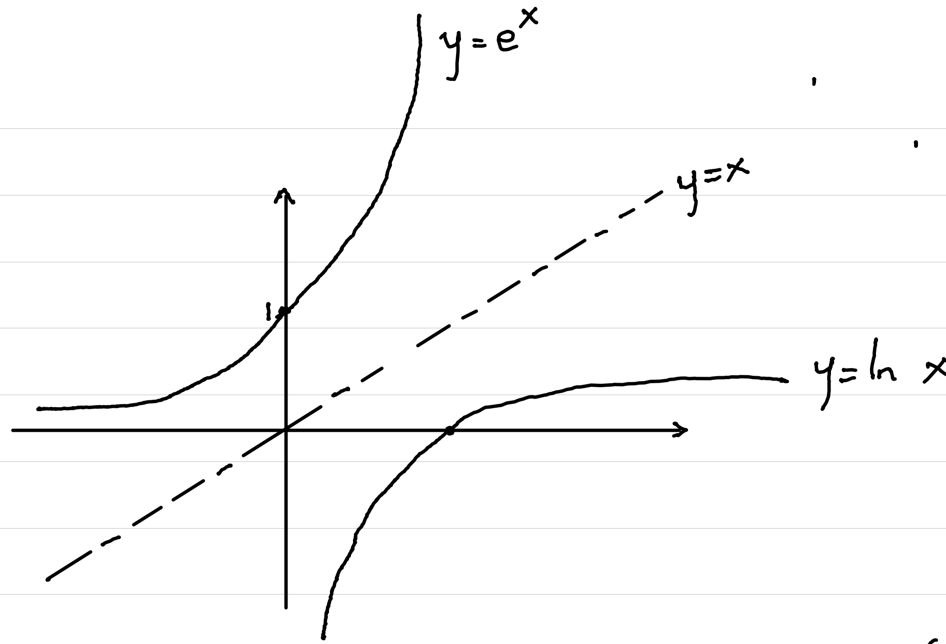
✓ We finally notice that ln and exp are inverses of each other.

$y = e^x$

✓ We can also notice that

$$\lim_{x\to\infty} e^x = \infty \quad \text{and} \quad \lim_{x\to-\infty} e^x = 0$$

✓ Hence,



$\frac{d}{dx}(e^{x^2}) = e^{x^2} \cdot 2x$

Derivatives.

✓ We can now have it that

$*$ $\frac{d}{dx}(e^x) = e^x$

$\frac{d}{dx}(a^x) = a^x \ln a$ where $a=e$.

$***$ $\frac{d}{dx}(e^u) = e^u \frac{du}{dx}$ where $u=g(x)$.

$\frac{d}{dx}(\log_a x) = \frac{1}{x \ln a}$

$*$ $\frac{d}{dx}(\ln x) = \frac{1}{x}$

$***$ $\frac{d}{dx}(\ln|u|) = \frac{1}{u}\frac{du}{dx}$ where $u=g(x)$.

"Proof"

- Recall that

$$|u| = \begin{cases} u, & u \geq 0 \\ -u, & u < 0 \end{cases}$$

- Hence, we can do $***$ in 2 steps.

① If $u>0$, $\ln|u| = \ln u$.

Hence,

$$\frac{d}{dx}(\ln|u|) = \frac{d}{dx}(\ln u)$$
$$= \frac{1}{u}\frac{du}{dx}$$

② If $u<0$, $\ln|u| = \ln(-u)$.

Hence,

$$\frac{d}{dx}(\ln|u|) = \frac{d}{dx}(\ln(-u))$$
$$= \frac{1}{-u} \cdot \frac{d}{dx}(-u)$$
$$= \frac{1}{u}\frac{du}{dx}$$

Example 1

Find $\frac{dy}{dx}$ if $y = \ln(x^2+1)$.

Sol^n

- Let $u = x^2+1$. Then $y = \ln u$.
- Hence,

$$\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx} \quad \text{(chain rule)}$$

$$= \frac{1}{u} \cdot 2x$$

$$= \frac{2x}{u}$$

$$= \frac{2x}{x^2+1} \quad \text{since } u = x^2+1.$$

Example 2

Find $\frac{dy}{dx}$ if $y = x^3 e^{\sqrt{x+1}}$.

(labels: u over x^3, v over $e^{\sqrt{x+1}}$)

<u>Method 1</u> (Product rule)

Let $u = x^3$ and $v = e^{\sqrt{x+1}}$.

Then,

$$\frac{du}{dx} = 3x^2 \quad \text{and} \quad \frac{dv}{dx} = e^{\sqrt{x+1}} \cdot \frac{d}{dx}\left(\sqrt{x+1}\right)$$

$$= \frac{e^{\sqrt{x+1}}}{2\sqrt{x+1}} \quad \text{Check!! :)}$$

Hence,

$$\frac{dy}{dx} = \frac{du}{dx} \cdot v + u \cdot \frac{dv}{dx}$$

$$= 3x^2 e^{\sqrt{x+1}} + \frac{x^3 e^{\sqrt{x+1}}}{2\sqrt{x+1}} = \quad ??$$

Method 2 (Implicit).